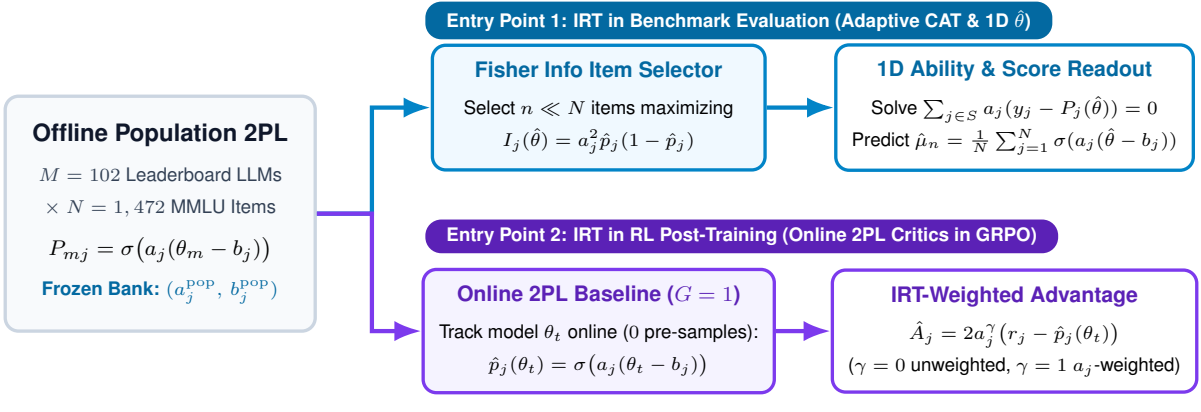

Offline Population 2PL
$M = 102$ Leaderboard LLMs
$\times N = 1,472$ MMLU Items
$P_{mj} = \sigma(a_j(\theta_m - b_j))$
Frozen Bank: $(a_j^{\text{pop}}, b_j^{\text{pop}})$
Entry Point 1: IRT in Benchmark Evaluation (Adaptive CAT & 1D $\hat{\theta}$)
Fisher Info Item Selector
Select $n \ll N$ items maximizing
$I_j(\hat{\theta}) = a_j^2 \hat{p}_j(1 - \hat{p}_j)$
1D Ability & Score Readout
Solve $\sum_{j \in S} a_j(y_j - P_j(\hat{\theta})) = 0$
Predict $\hat{\mu}_n = \frac{1}{N} \sum_{j=1}^{N} \sigma(a_j(\hat{\theta} - b_j))$
Entry Point 2: IRT in RL Post-Training (Online 2PL Critics in GRPO)
Online 2PL Baseline ($G = 1$)
Track model θ_t online (0 pre-samples):
$\hat{p}_j(\theta_t) = \sigma(a_j(\theta_t - b_j))$
IRT-Weighted Advantage
$\hat{A}_j = 2a_j^{\gamma}(r_j - \hat{p}_j(\theta_t))$
($\gamma = 0$ unweighted, $\gamma = 1$ a_j-weighted)